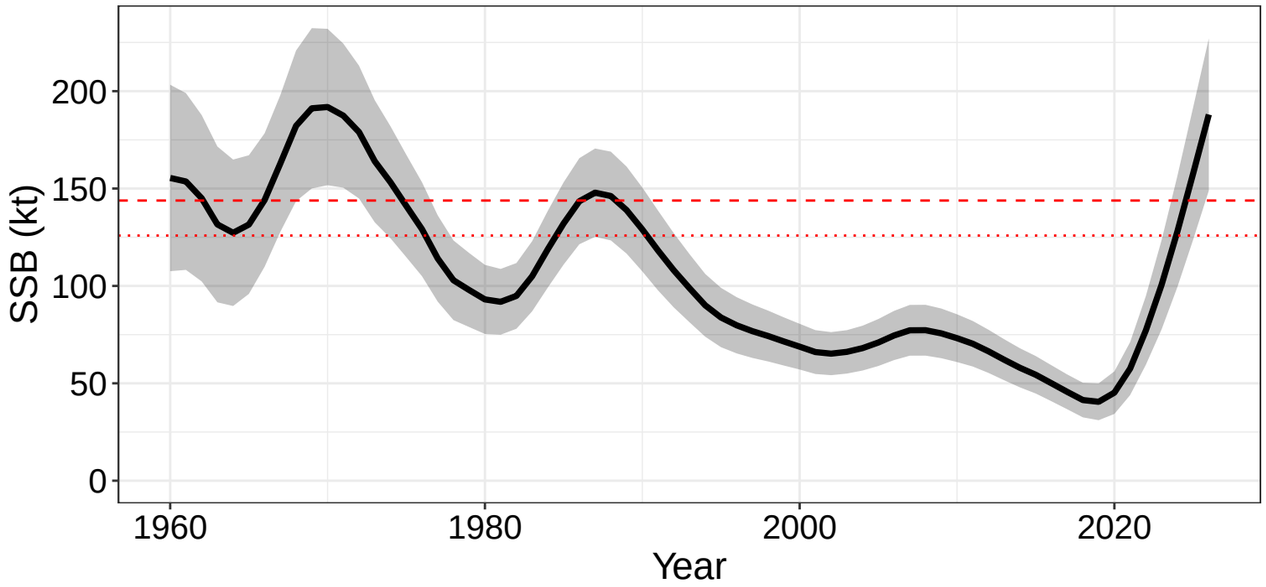
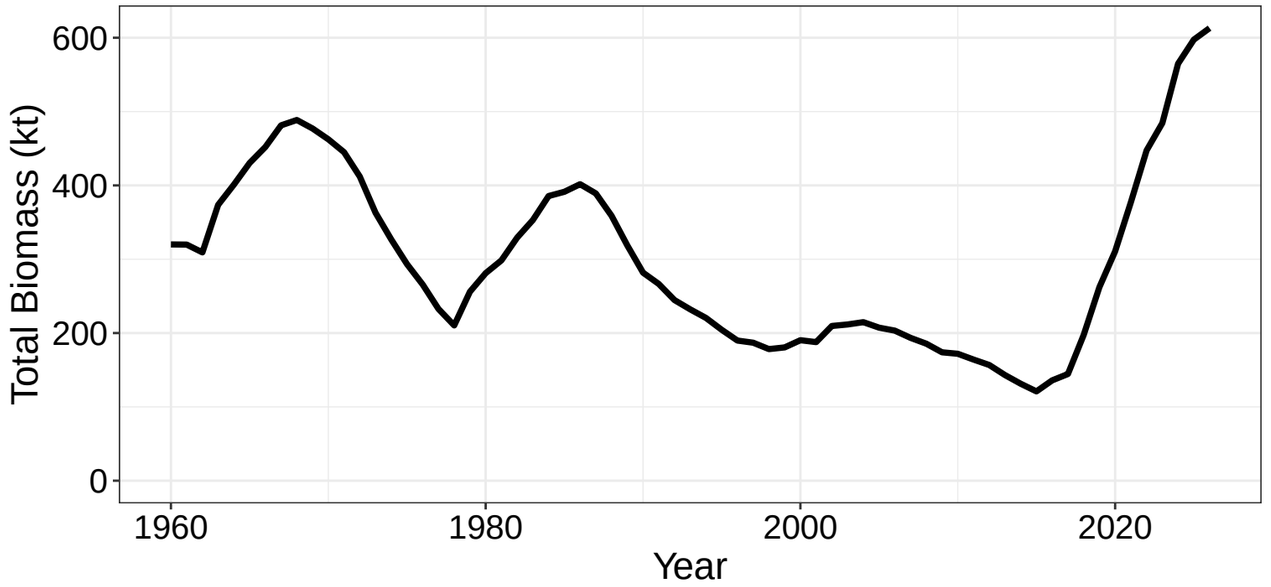
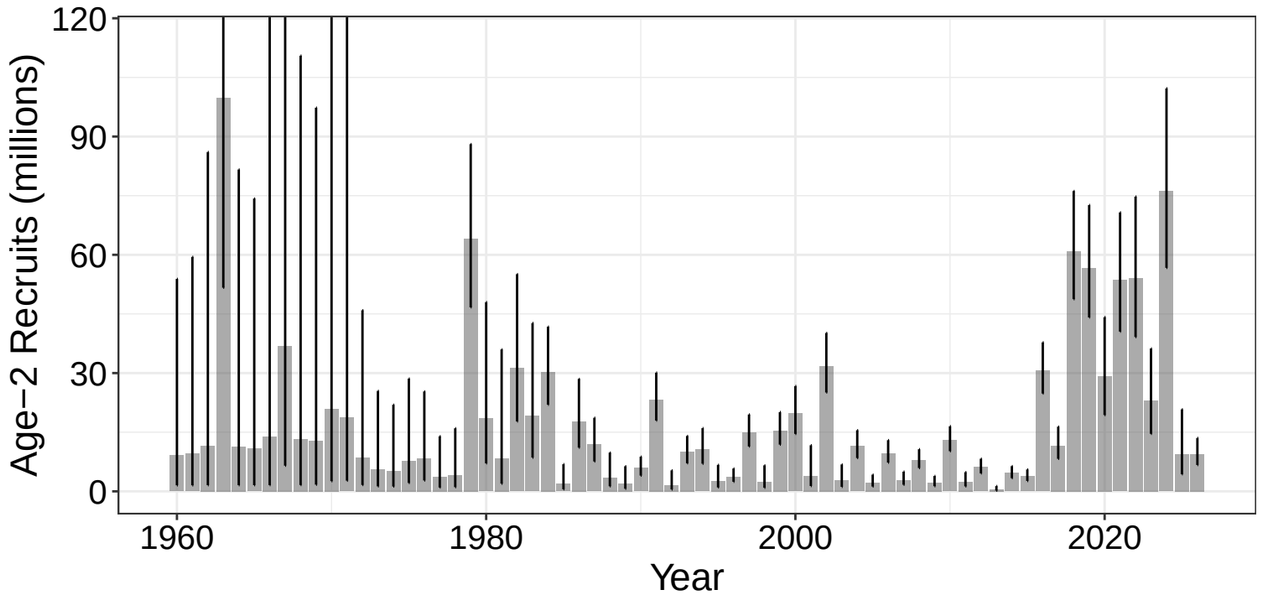
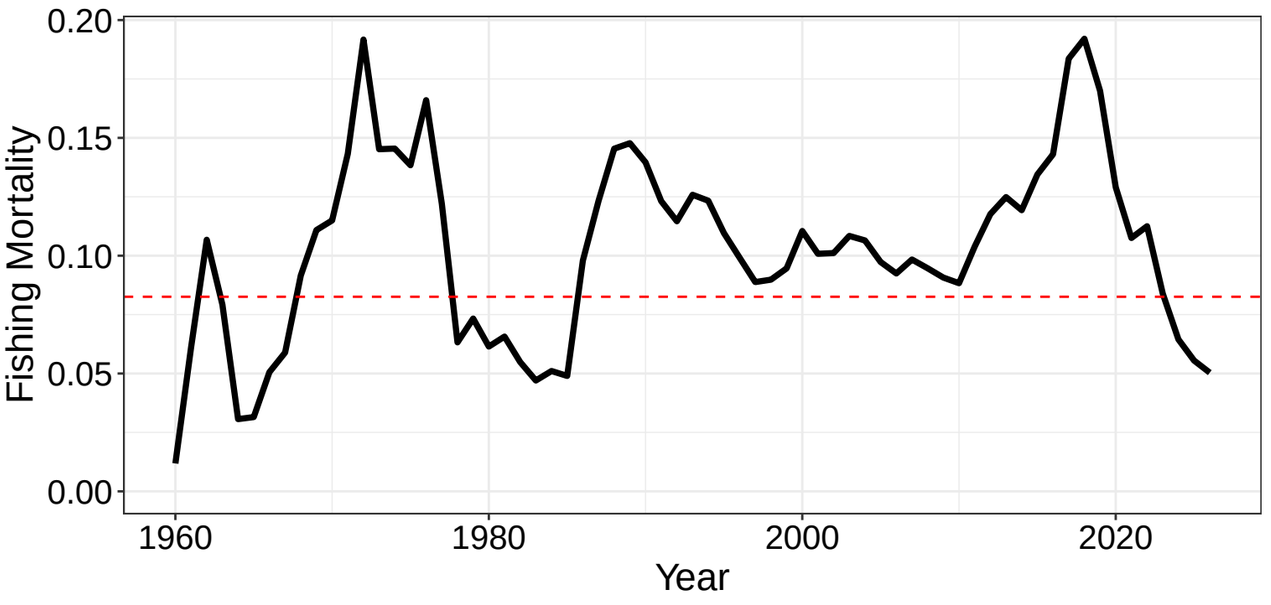
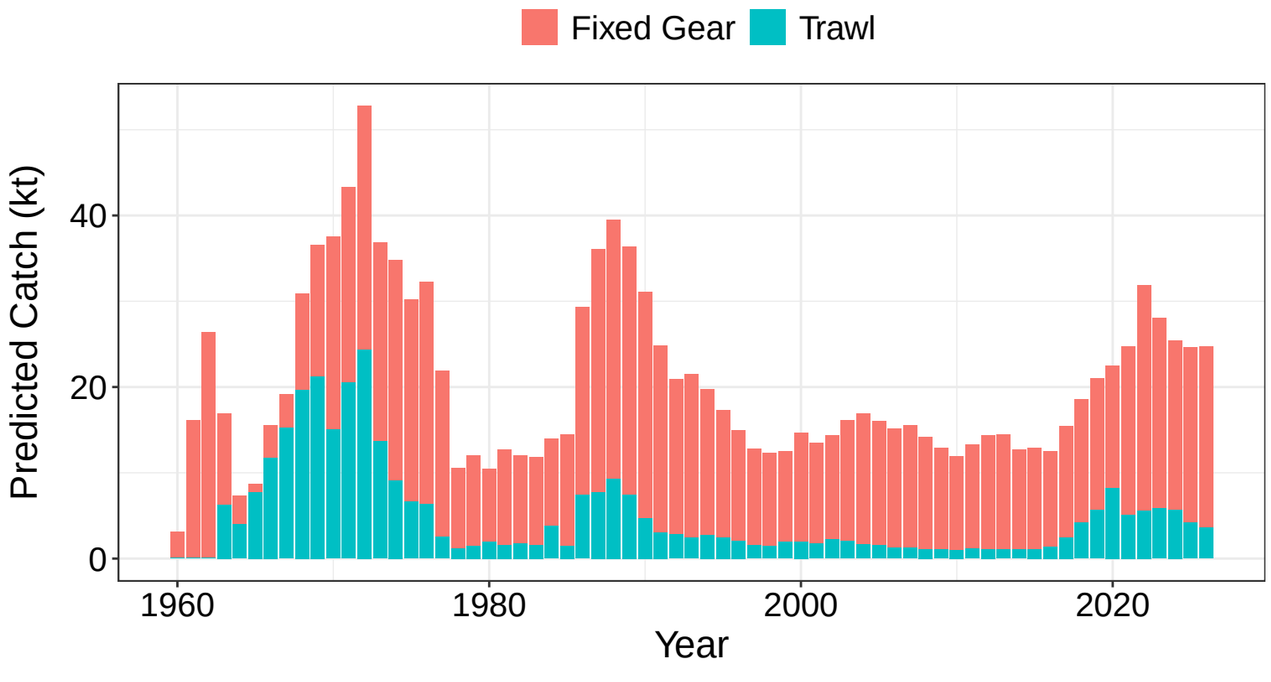
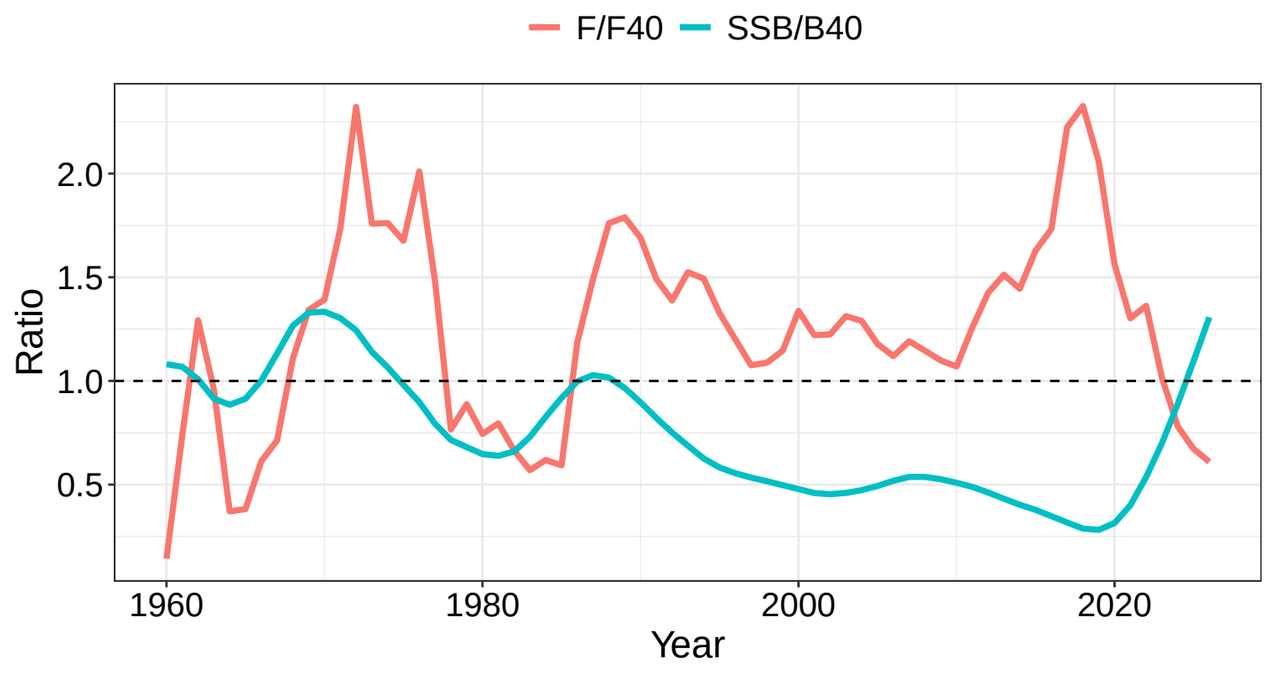

SSB (kt)
200
150
100
50
0
1960
1980
2000
2020
Year

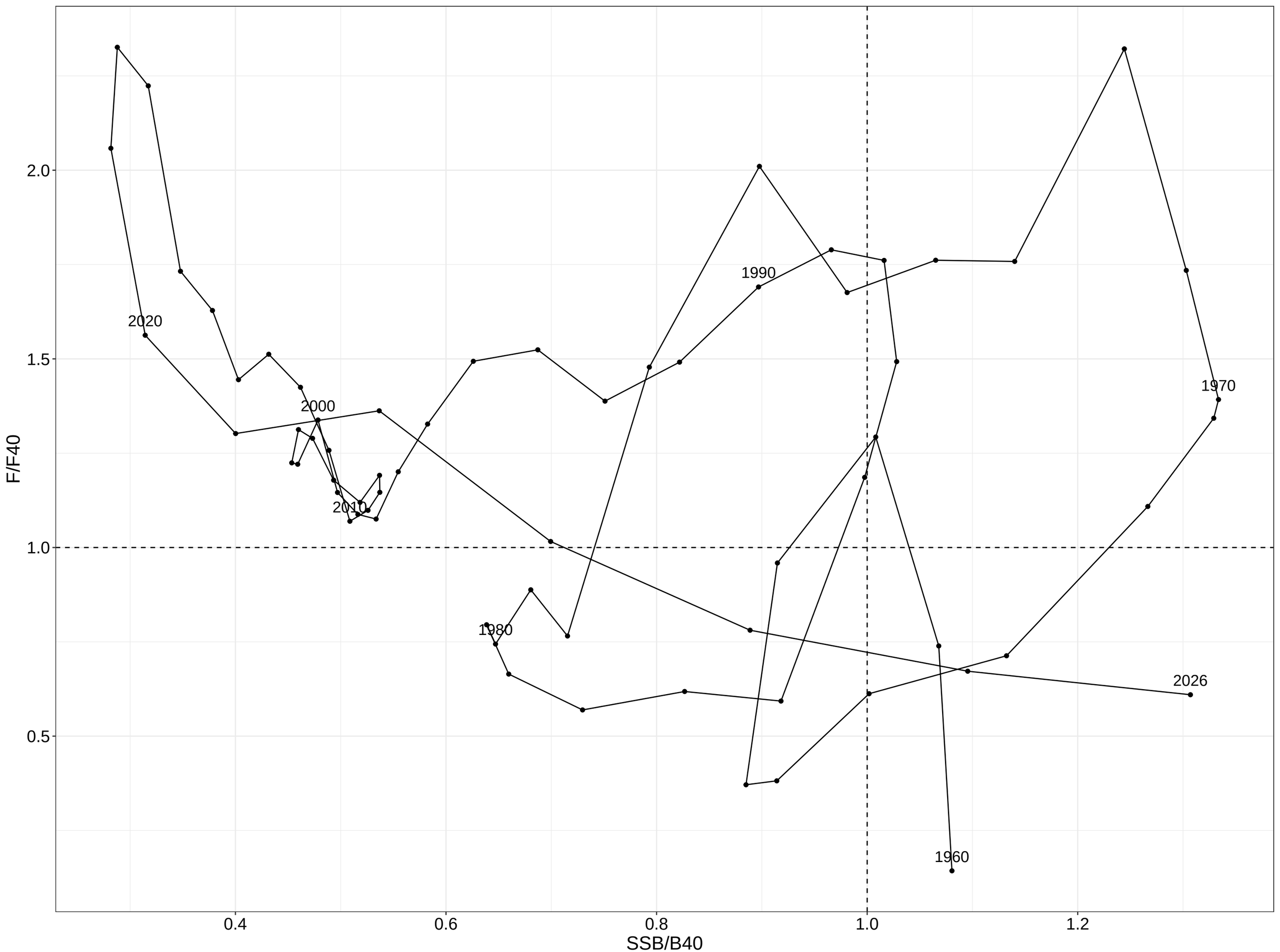

Total Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year


Age−2 Recruits (millions)
120
90
60
30
0
1960
1980
2000
2020
Year


Fishing Mortality
0.20
0.15
0.10
0.05
0.00
1960
1980
2000
2020
Year


Fixed Gear
Trawl
Predicted Catch (kt)
40
20
0
1960
1980
2000
2020
Year


F/F40
SSB/B40
Ratio
2.0
1.5
1.0
0.5
1960
1980
2000
2020
Year


F/F40
SSB/B40
2.0
1.5
1.0
0.5
0.4
0.6
0.8
1.0
1.2
2020
2000
2010
1990
1970
1980
2026
1960

Domestic LL Survey RPW (not fit)

Japanese LL Survey RPW (not fit)

Domestic LL Survey RPN

Japanese LL Survey RPN

Domestic LL Fishery CPUE

Japanese LL Fishery CPUE

GOA Trawl Survey Biomass

Index

Year

Domestic LL Survey RPN

Japanese LL Survey RPN

Domestic LL Fishery CPUE

Japanese LL Fishery CPUE

GOA Trawl Survey Biomass

log(obs/pred)

Year

Female
Male
Fixed Gear Fishery (pre−IFQ)
Fixed Gear Fishery (IFQ)
Trawl Fishery
Domestic LL Survey
Japanese LL Survey
GOA Trawl Survey
Selectivity
1.00
0.75
0.50
0.25
0.00
10
20
30
Age

Fixed Gear Fishery

Domestic LL Survey

Japanese LL Survey

Proportion

0.00

0.05

0.10

10

20

30

Age

Fixed Gear Fishery Female
Fixed Gear Fishery Male
Trawl Fishery Female
Trawl Fishery Male
Domestic LL Survey Female
Domestic LL Survey Male
Japanese LL Survey Female
Japanese LL Survey Male
GOA Trawl Survey Female
GOA Trawl Survey Male
Proportion
Length (cm)

Negative Positive |Resid| 1 2 3 4

Fixed Gear Fishery

Domestic LL Survey

Japanese LL Survey

Age

Year

Negative Positive |Resid| 0 1 2 3 4

Fixed Gear Fishery Female

Fixed Gear Fishery Male

Trawl Fishery Female

Trawl Fishery Male

Domestic LL Survey Female

Domestic LL Survey Male

Japanese LL Survey Female

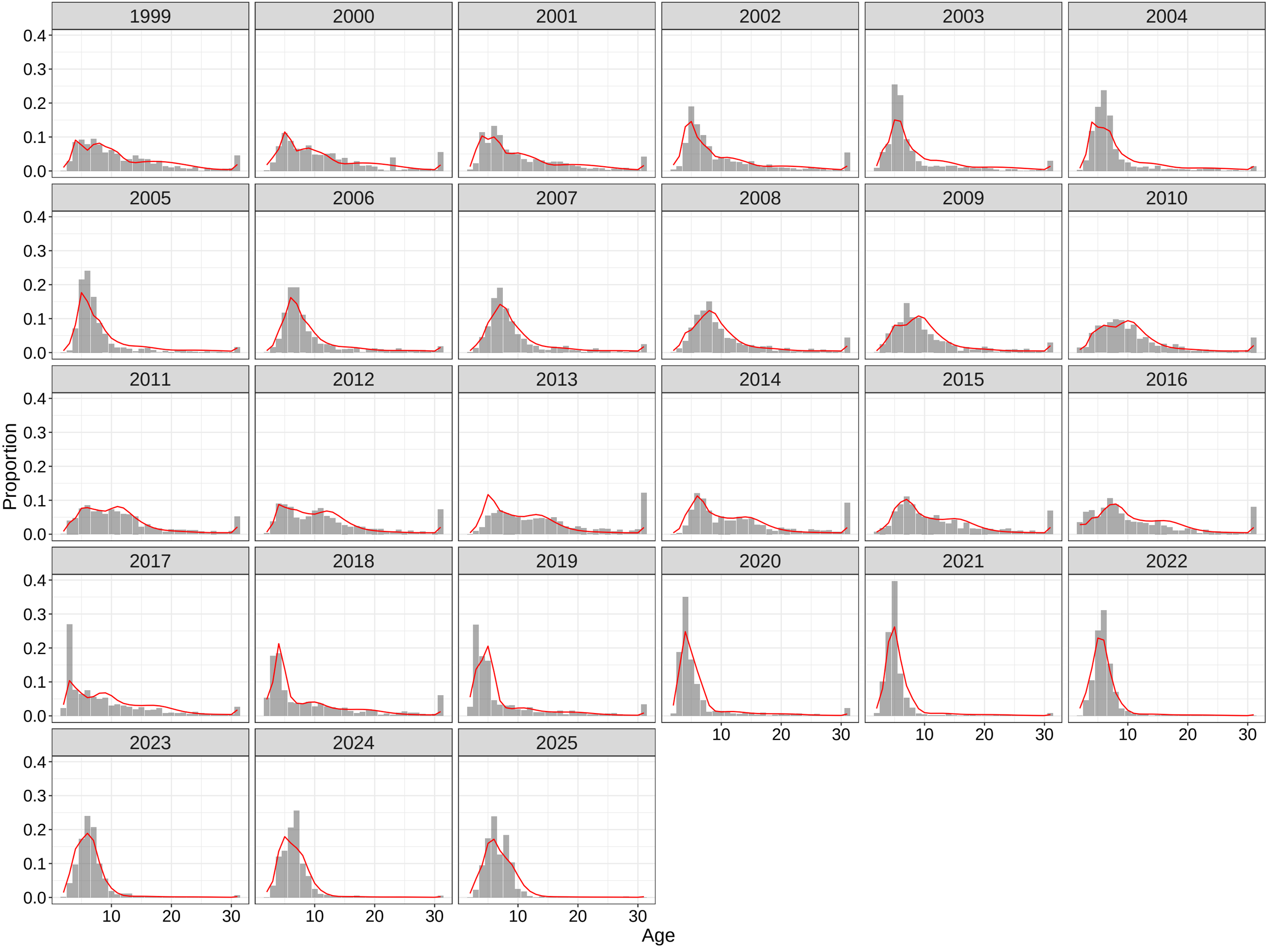
Japanese LL Survey Male

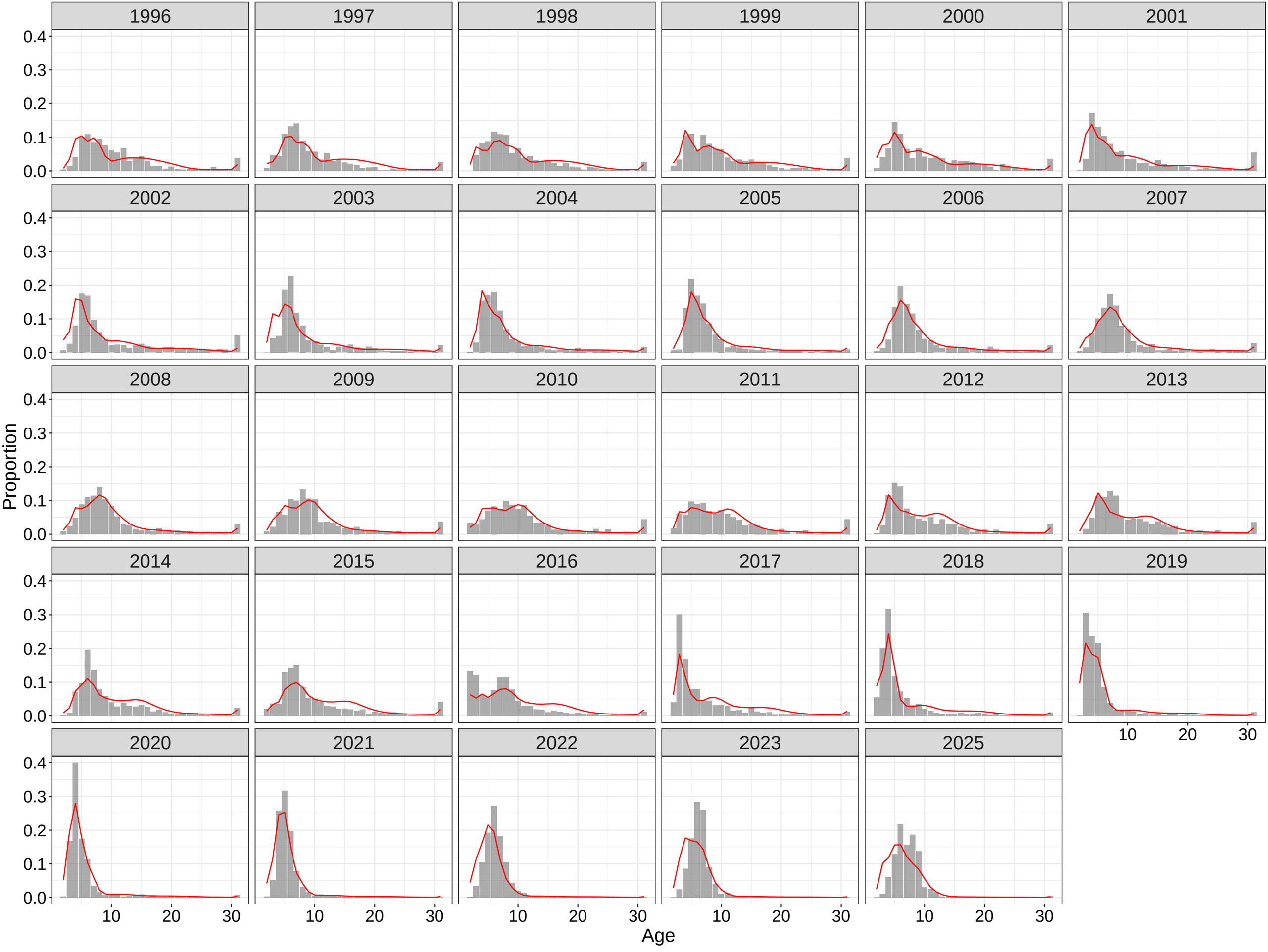
GOA Trawl Survey Female

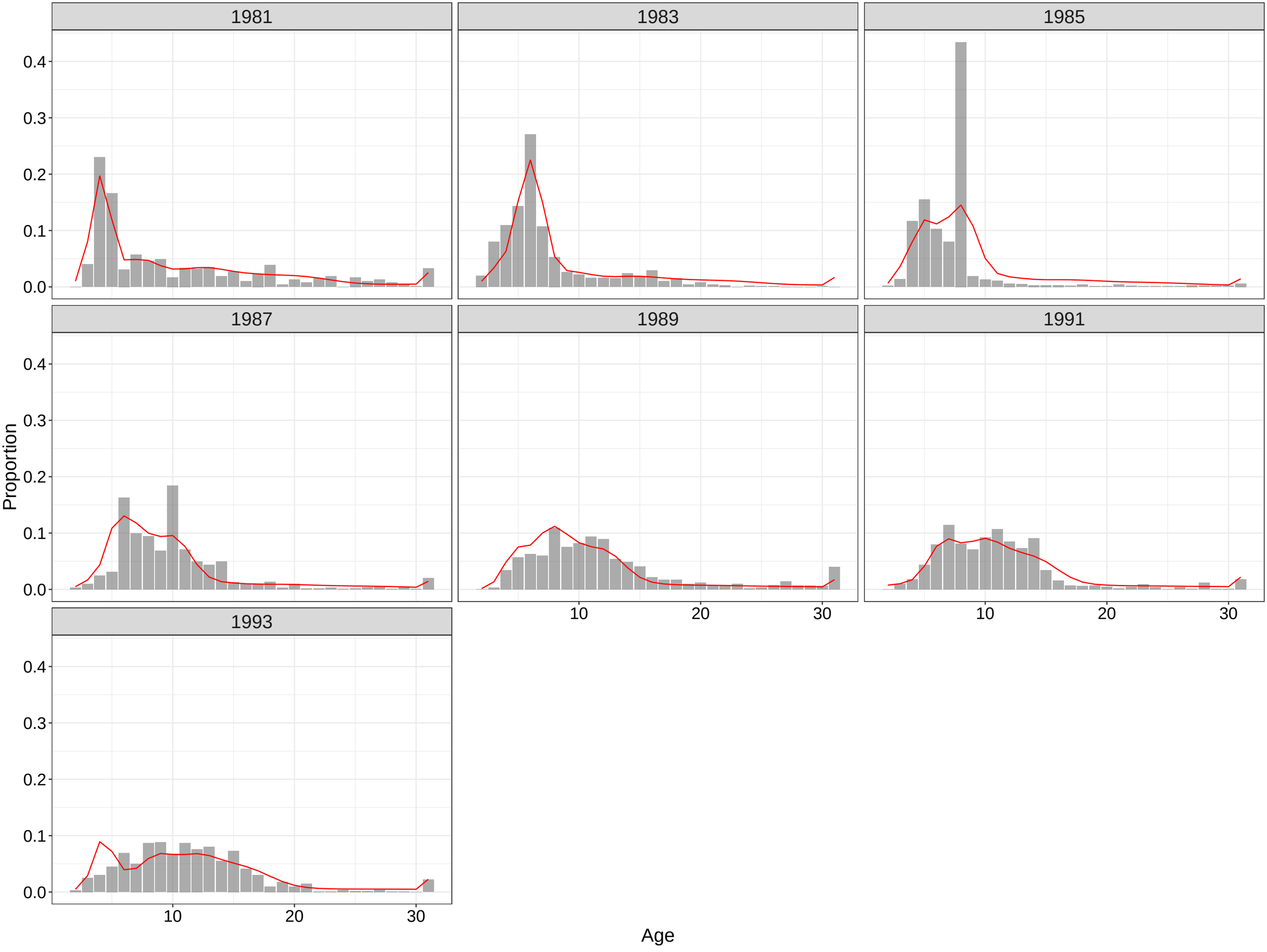
GOA Trawl Survey Male

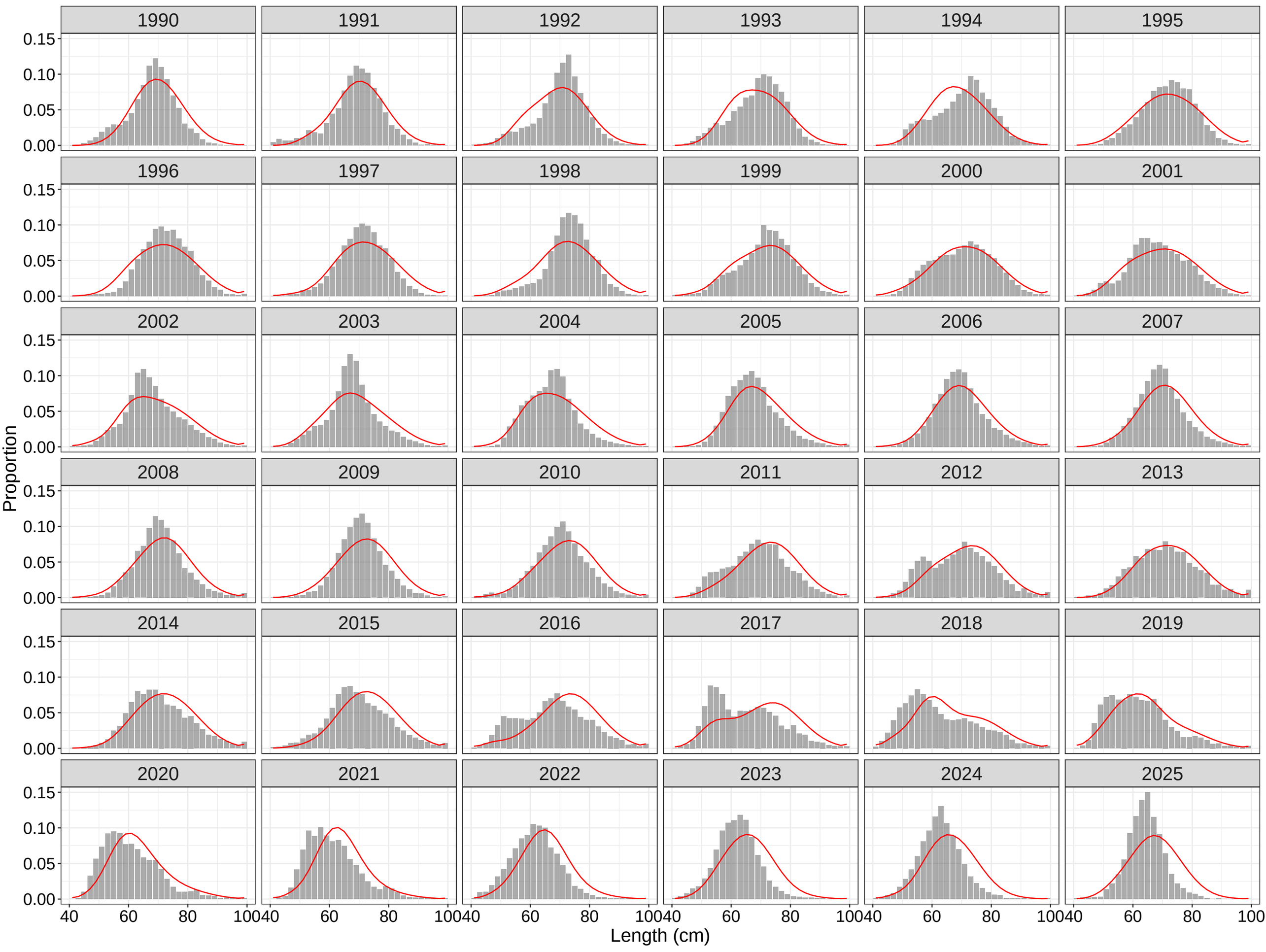
Length (cm)

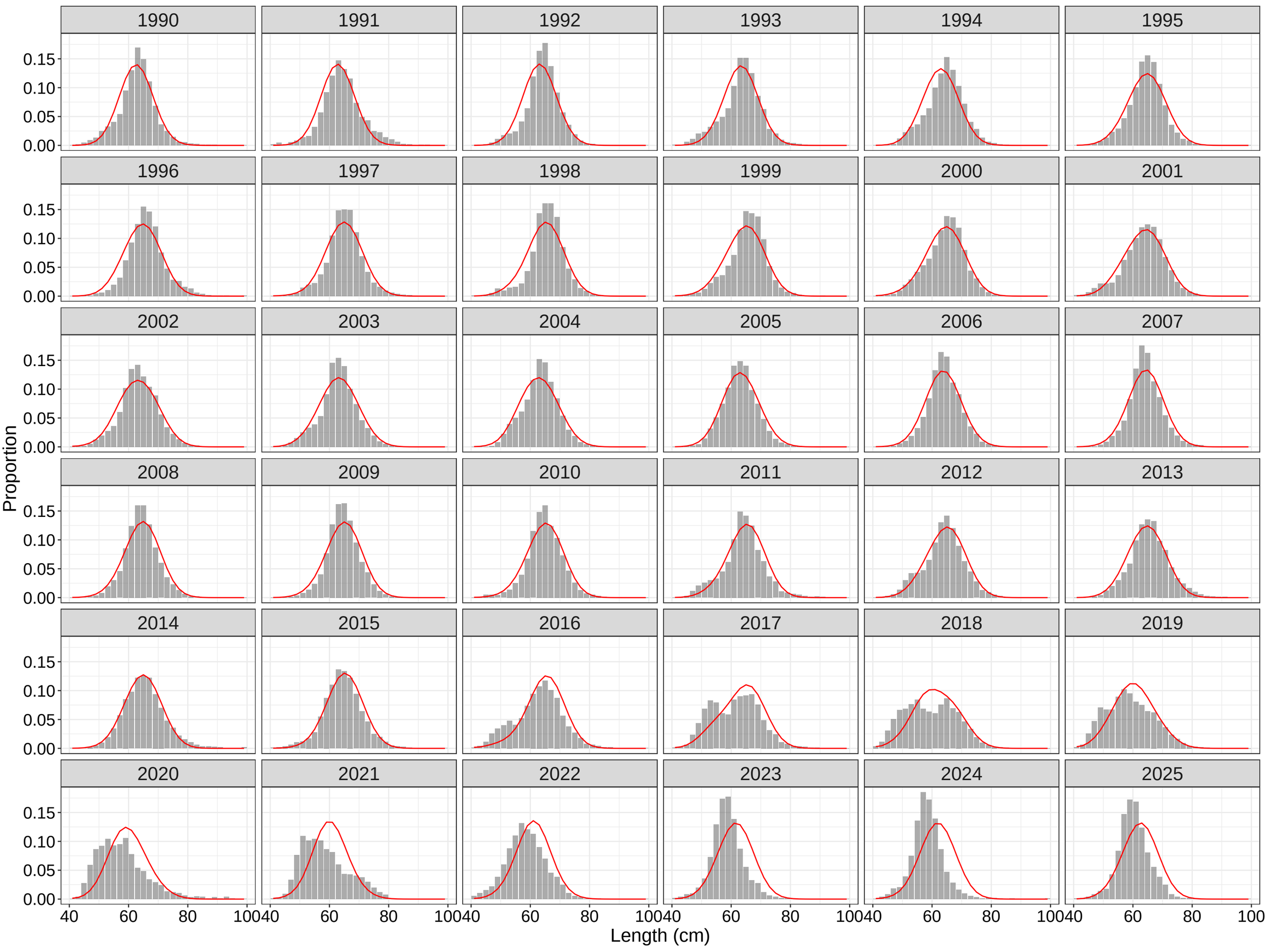
Year


1999
2000
2001
2002
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2004
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2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Proportion
0.0
0.1
0.2
0.3
0.4
10
20
30
Age

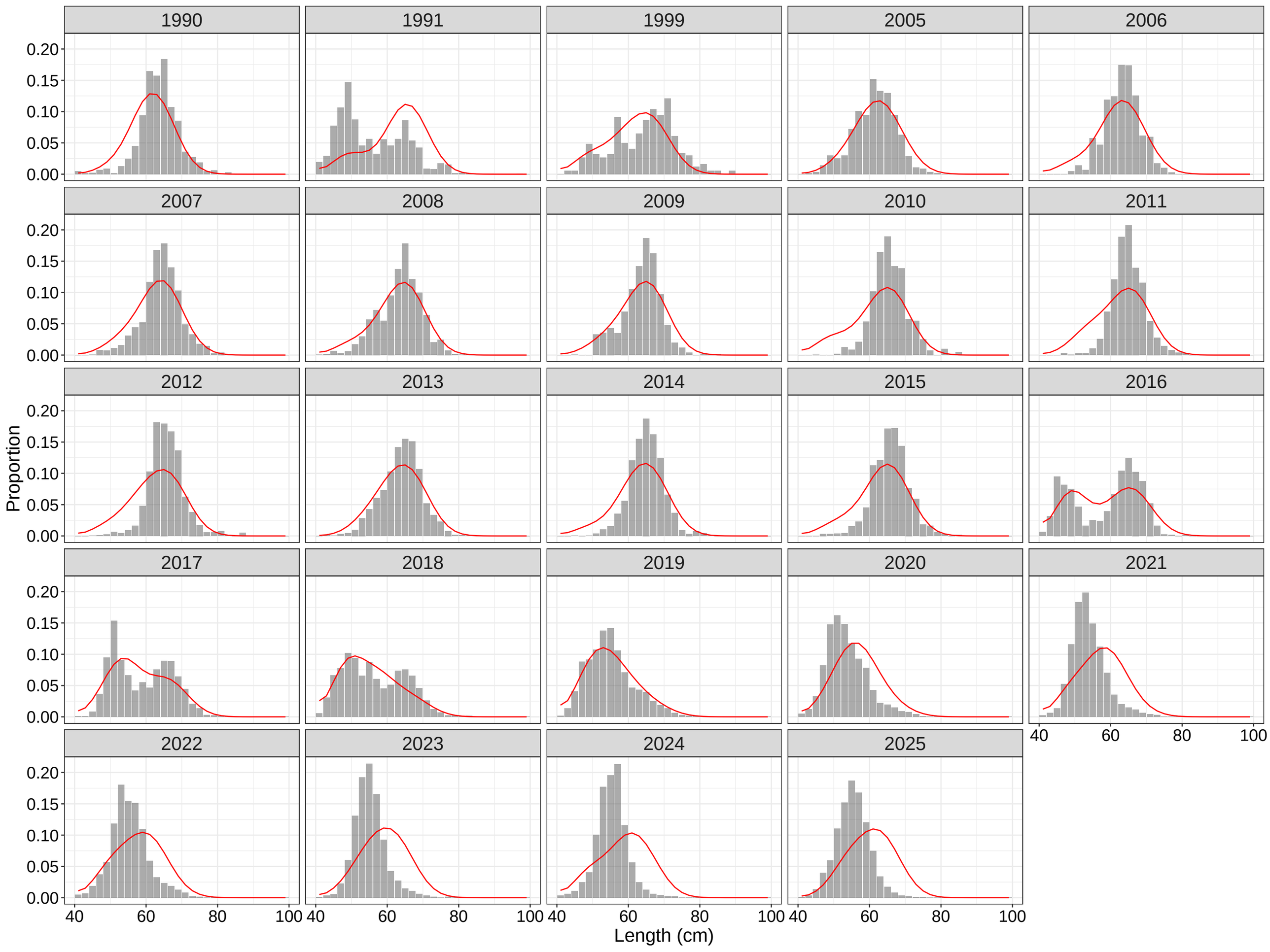

1996
1997
1998
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2006
2007
2008
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2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2025
Proportion
Age
0.0
0.1
0.2
0.3
0.4
10
20
30

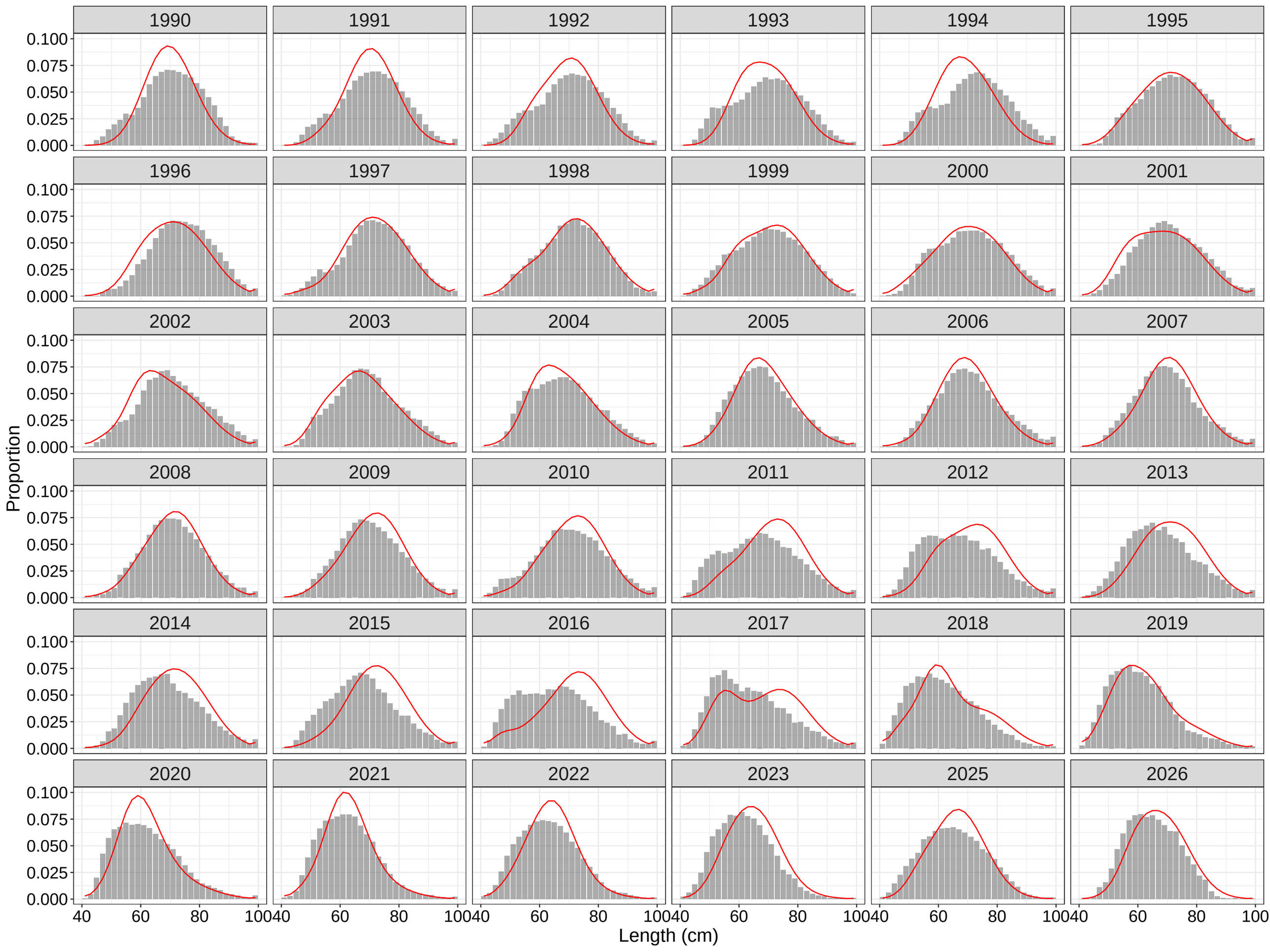

1981
1983
1985
1987
1989
1991
1993
Proportion
Age
0.0
0.1
0.2
0.3
0.4
10
20
30


1990
1991
1992
1993
1994
1995
1996
1997
1998
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2012
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2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
0.15
0.10
0.05
0.00
40
60
80
100
Proportion
Length (cm)


1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
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2018
2019
2020
2021
2022
2023
2024
2025
0.00
0.05
0.10
0.15
40
60
80
100
Proportion
Length (cm)

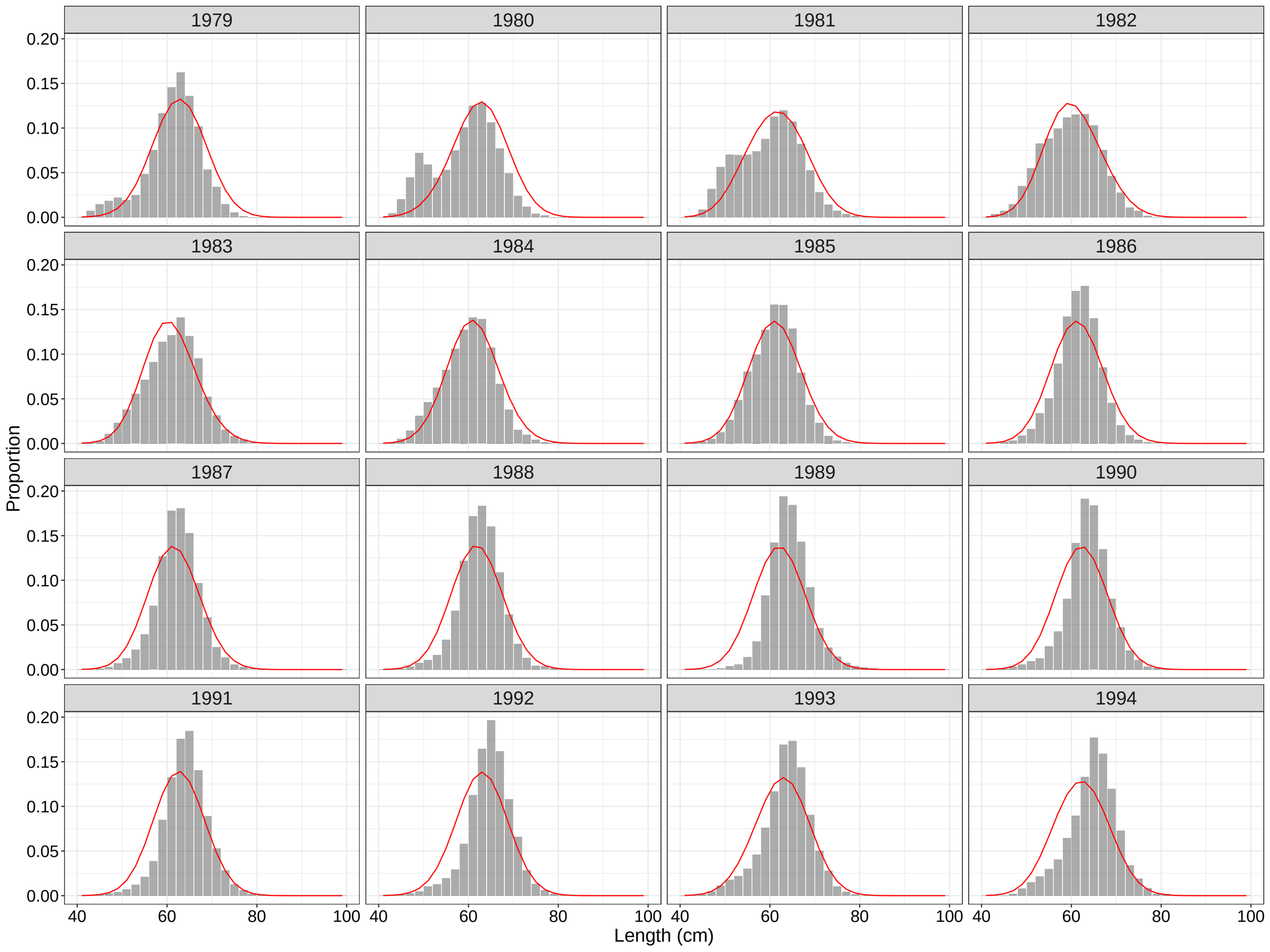
1990 1991 1999 2005 2006
2007 2008 2009 2010 2011
2012 2013 2014 2015 2016
2017 2018 2019 2020 2021
2022 2023 2024 2025
Proportion
0.00 0.05 0.10 0.15 0.20
40 60 80 100
Length (cm)

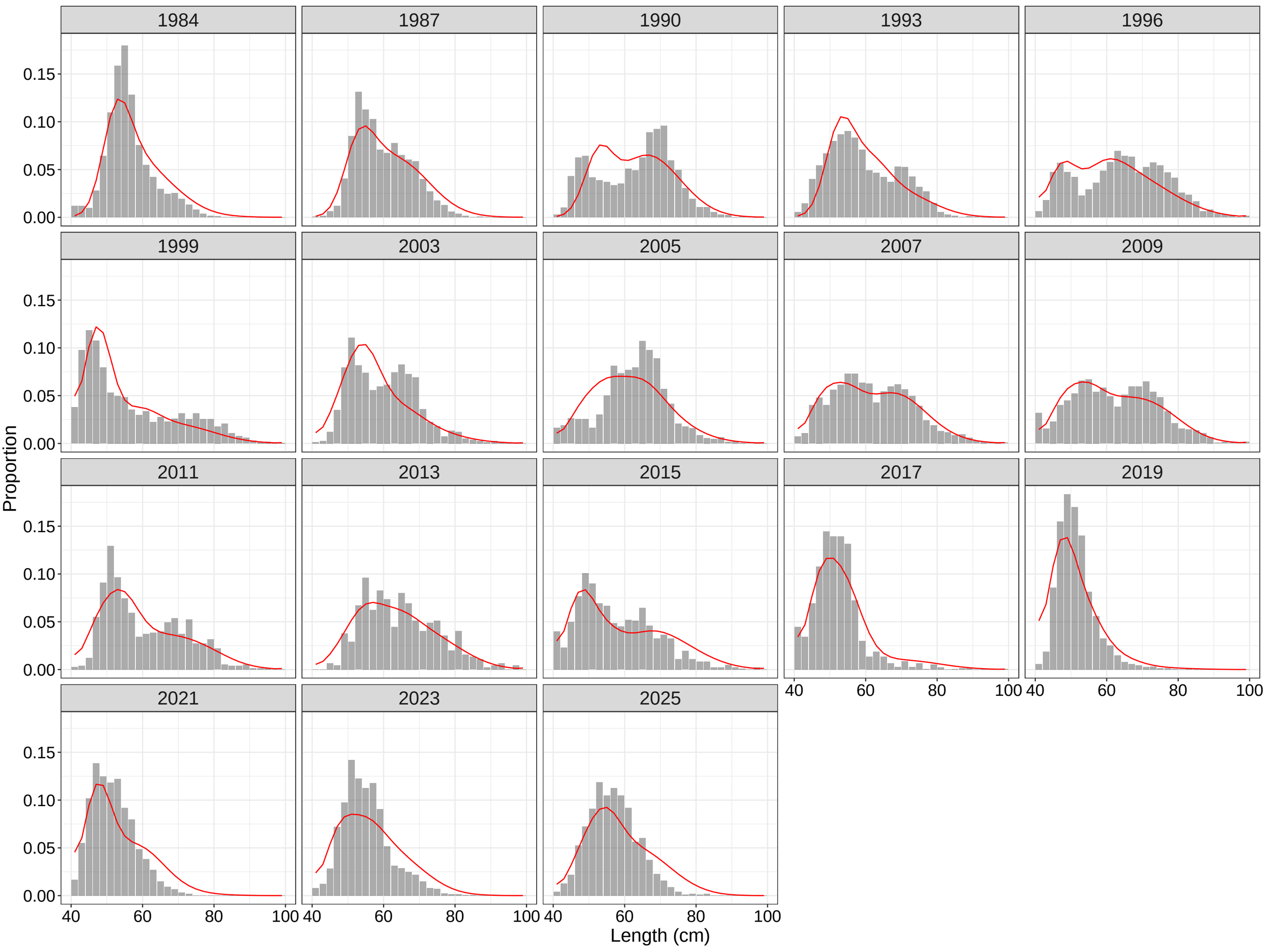

1990
1991
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2008
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2010
2011
2012
2013
2014
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2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Proportion
0.00
0.05
0.10
0.15
0.20
40
60
80
100
Length (cm)

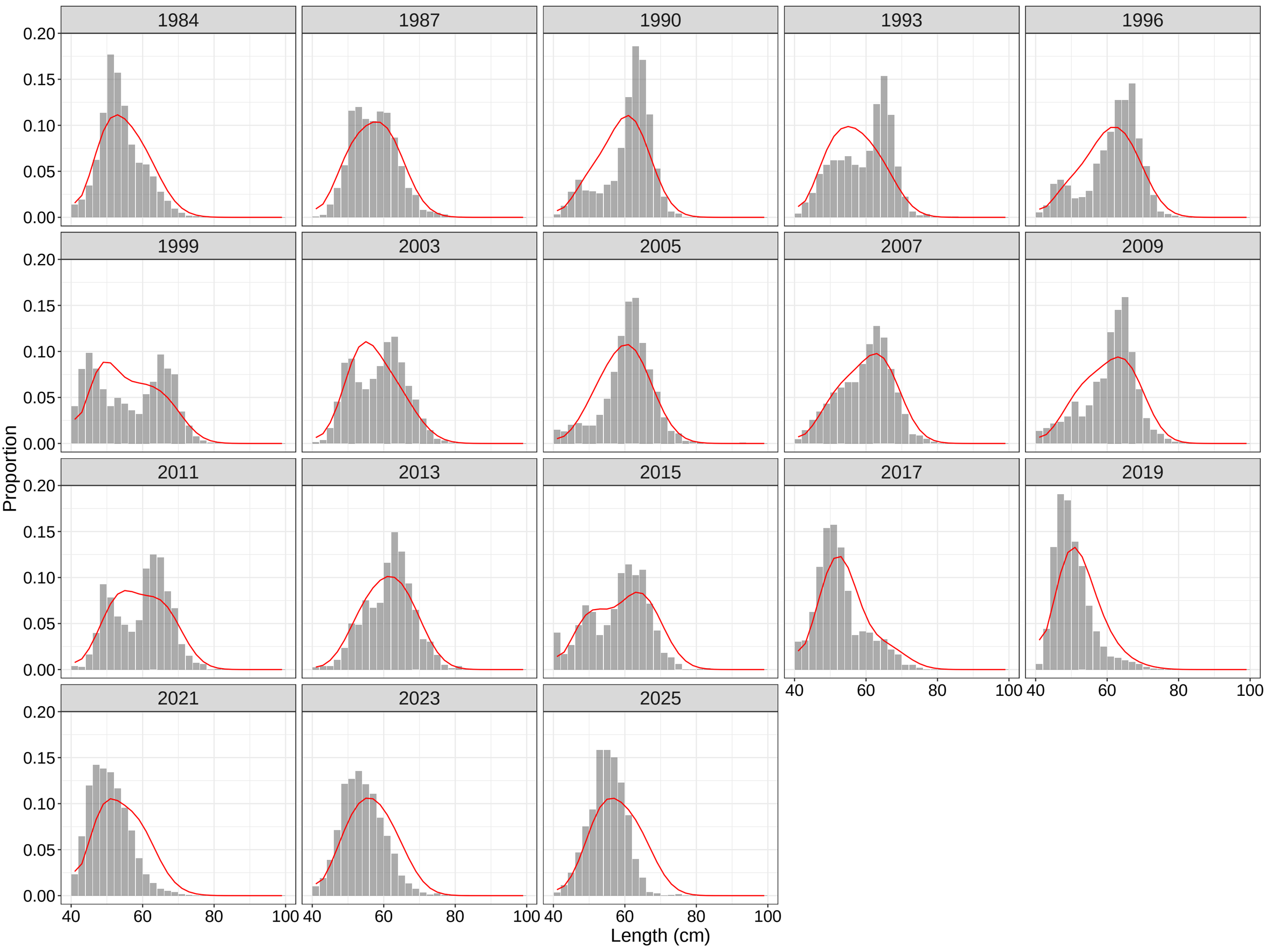

1990
1991
1992
1993
1994
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1996
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1998
1999
2000
2001
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2005
2006
2007
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2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2025
2026
0.100
0.075
0.050
0.025
0.000
Proportion
40
60
80
100
Length (cm)

1990 1991 1992 1993 1994 1995
1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025 2026
0.00 0.05 0.10
Proportion
40 60 80 100
Length (cm)

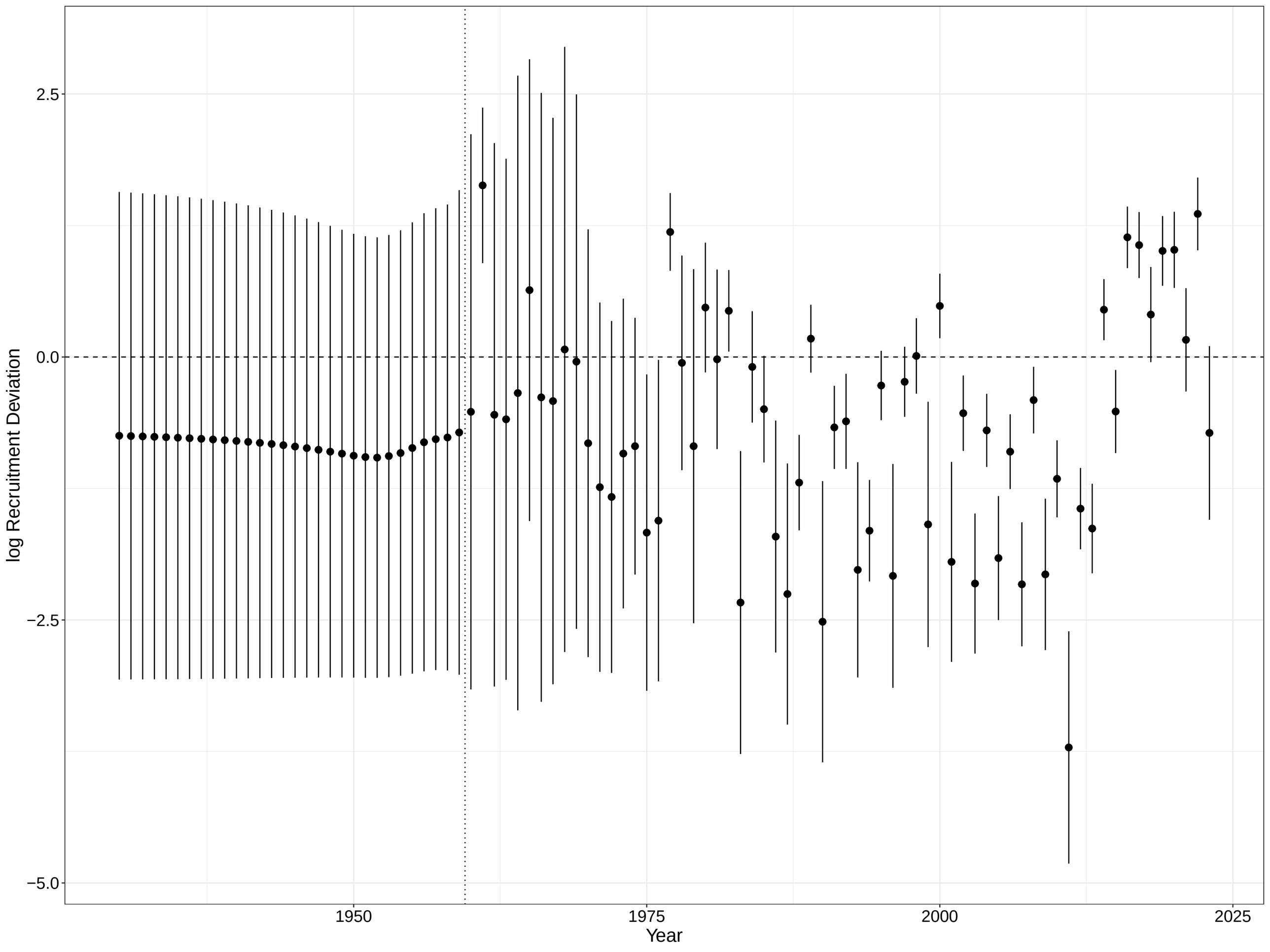
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994
0.00 0.05 0.10
40 60 80 100
Proportion
Length (cm)

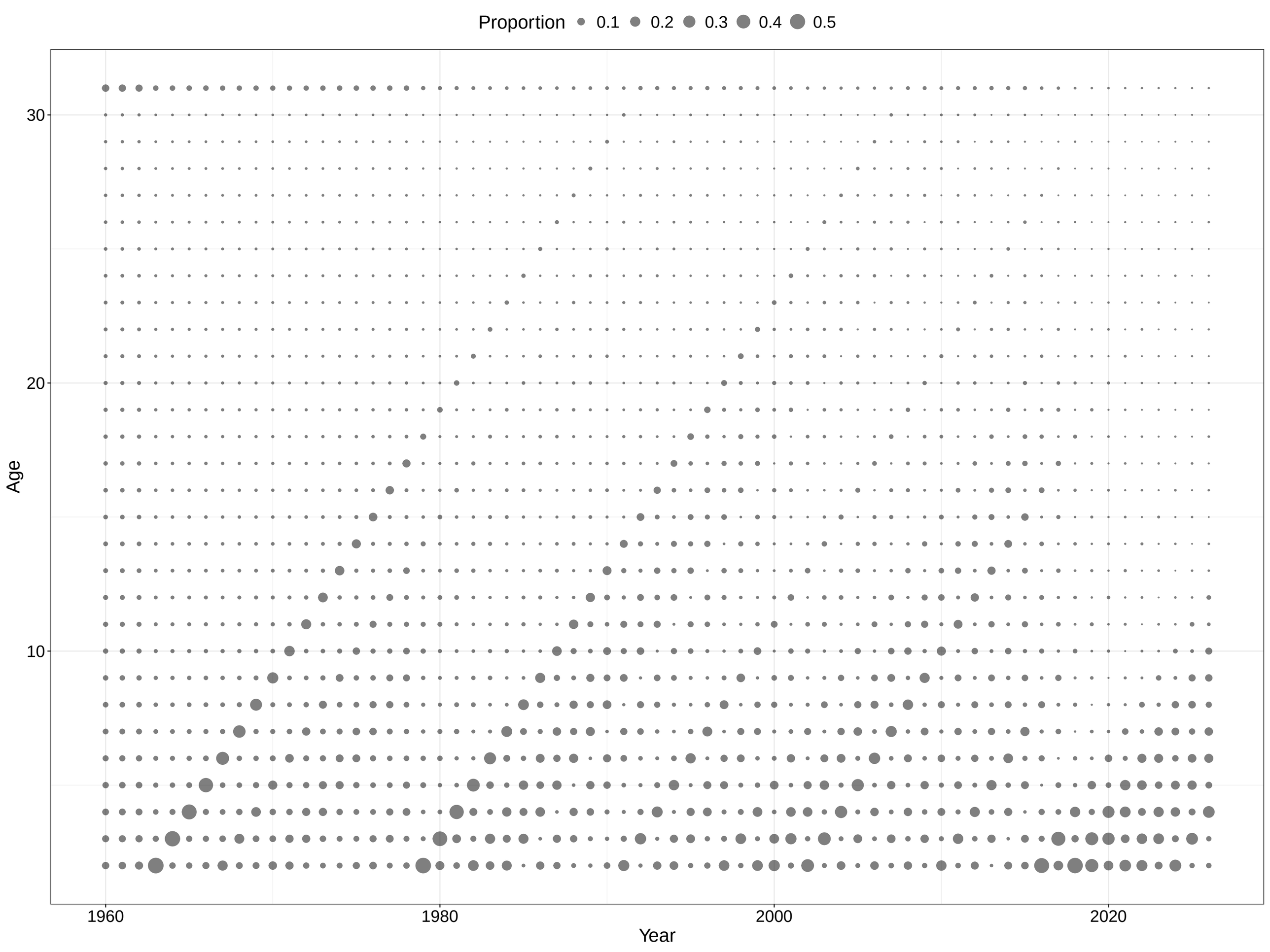

1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
0.20
0.15
0.10
0.05
0.00
Proportion
40
60
80
100
Length (cm)


1984
1987
1990
1993
1996
1999
2003
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
2025
0.15
0.10
0.05
0.00
40
60
80
100
Proportion
Length (cm)


1984
1987
1990
1993
1996
1999
2003
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
2025
0.20
0.15
0.10
0.05
0.00
40
60
80
100
Proportion
Length (cm)

Age Comps
Catch
Indices
Length Comps
Penalties
nLL
600
400
200
0
Fishery Size Composition Trawl
Survey Age Composition DomesticLL
Survey Size Composition GOATrawl
Fishery Age Composition
Domestic Survey Abundance Index
Survey Age Composition CooperativeLL
Survey Size Composition Domestic LL
Fishery Size Composition Fixed
Domestic LL Fishery CPUE
Survey Size Composition Cooperative LL
Recruitment Deviations
GOA Trawl Survey Biomass
SSQ Catch
Priors
Cooperative Survey Abundance Index
Fishing Mortality Regularity Penalty
Japanese LL Fishery CPUE


log Recruitment Deviation
2.5
0.0
−2.5
−5.0
1950
1975
2000
2025
Year


Proportion
0.1
0.2
0.3
0.4
0.5
Age
30
20
10
1960
1980
2000
2020
Year

Female
Male
Weight (kg)
5
4
3
2
1
10
20
30
Age
Maturity x Weight (Female)
1.00
0.75
0.50
0.25
0.00
10
20
30
Age

Catch at maxABC (kt)

SSB (kt)

50
40
30
20
10
0

300
200
100
0

2000
2010
2020
2030
2040

Year

Model 16.5 26.1 26.2

SSB (kt)

300
200
100
0

Total Biomass (kt)

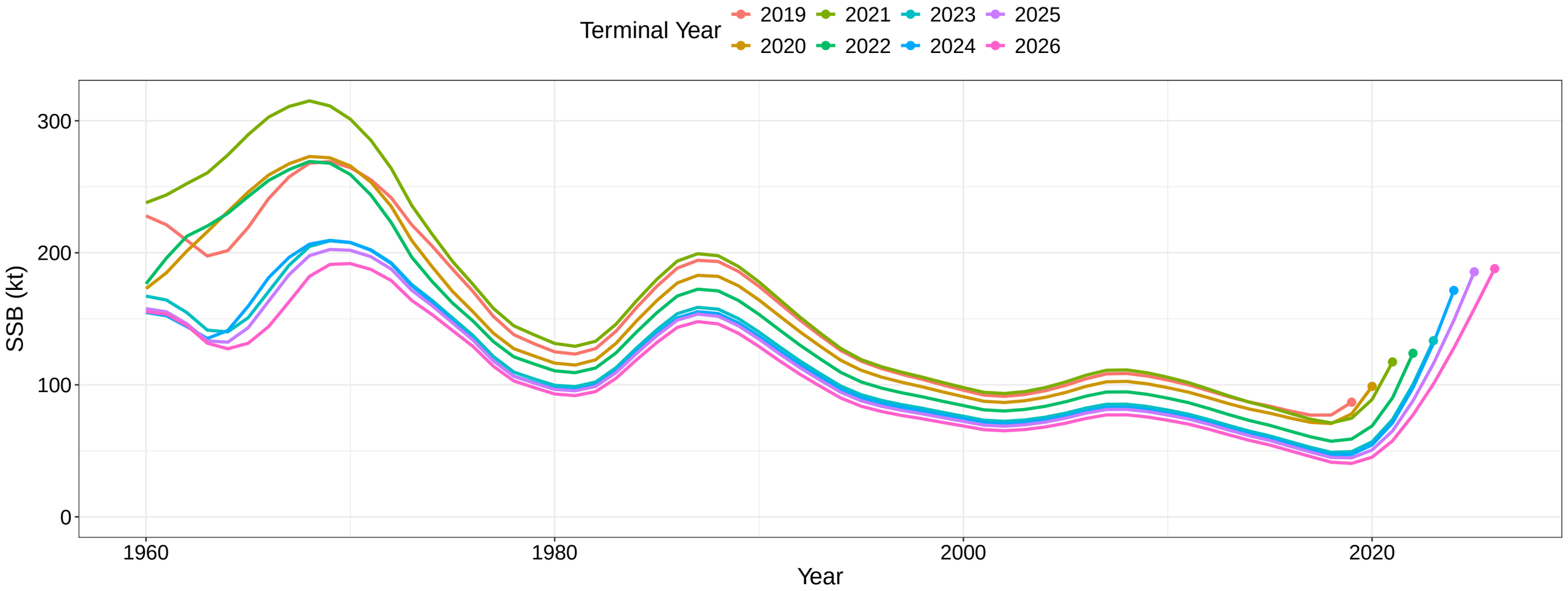
600
400
200
0

Age−2 Recruits (millions)

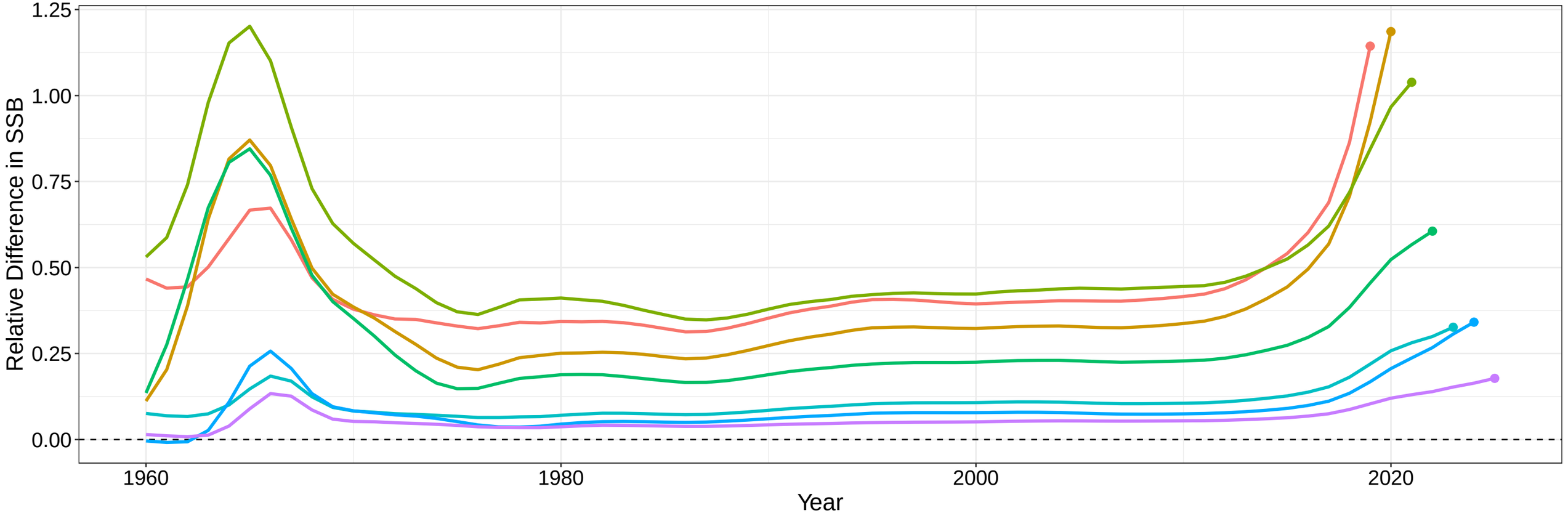
100
75
50
25
0

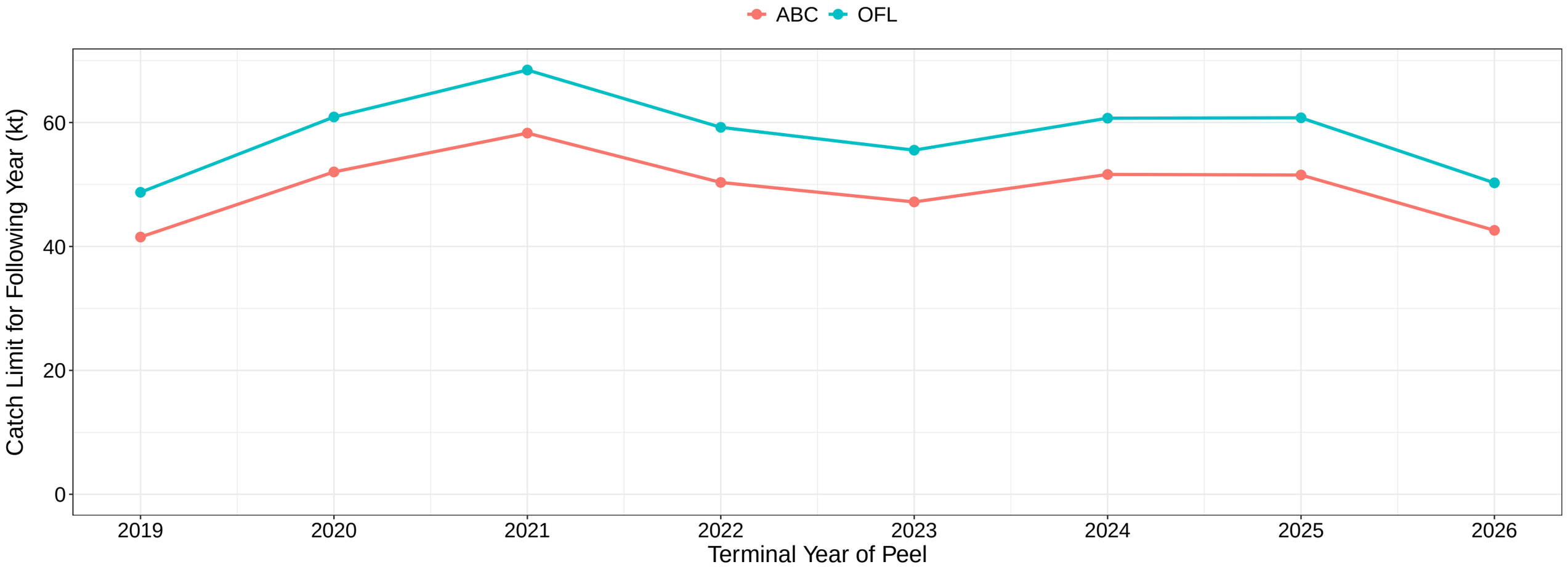
1960
1980
2000
2020

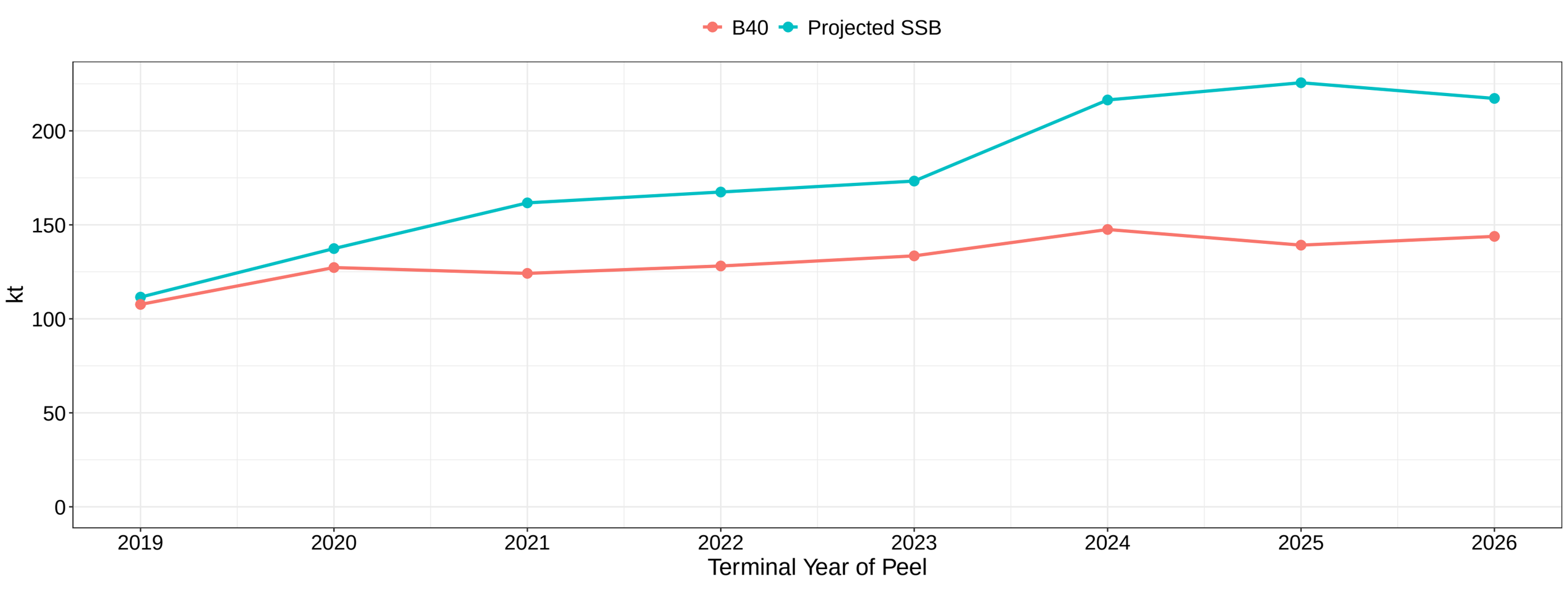
Year




Mohn's rho = 0.689




ABC
OFL
Catch Limit for Following Year (kt)
0
20
40
60
2019
2020
2021
2022
2023
2024
2025
2026
Terminal Year of Peel


B40
Projected SSB
kt
0
50
100
150
200
2019
2020
2021
2022
2023
2024
2025
2026
Terminal Year of Peel

Terminal Year
2019 2020 2021 2022 2023 2024 2025 2026

SSB/B40

2.5
2.0
1.5
1.0
0.5

1960 1980 2000 2020

Year

F/F40

2.0
1.5
1.0
0.5
0.0

1960 1980 2000 2020

Year